Fine-Tuned Large Language Model-Enabled Personalized Mental Health Intervention System for University of Sri Jayewardenepura Students Using DASS-21 Scores and Lifestyle Data

Thesis Report

Group 07

By:- Hanan M.H, Hettiarachchi I.S, Kalubovila S.A.

Department of Information Communication Technology

Faculty of Technology

University of Sri Jayewardenepura

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Research Students:

Name	Focus Area	Department	Faculty	University	Index Number	Contact Details
Hanan M.H	Software Technology	Information Communication Technology	Technology	University of Sri Jayewardenepura	ICT/21/847	ict21847@fot.sjp.ac.lk
Hettiarachchi I.S.	Software Technology	Information Communication Technology	Technology	University of Sri Jayewardenepura	ICT/21/848	ict21848@fot.sjp.ac.lk
Kalubovila S.A.	Software Technology	Information Communication Technology	Technology	University of Sri Jayewardenepura	ICT/21/864	ict21864@fot.sjp.ac.lk

Main Supervisor:

Mrs. Sankani Heenkenda

Lecturer

Faculty of Technology,

University of Sri Jayewardenepura

Email: sankaniheenkenda@sjp.ac.lk

Declaration

I certify that this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due reference is made in the text.

Acknowledgement

We wish to express our sincere gratitude to all individuals and entities who have contributed to the development of this research proposal, titled "Fine-Tuned Large Language Model-Enabled Personalized Mental Health Intervention System for University of Sri Jayewardenepura Students Using DASS-21 Scores and Lifestyle Data," completed on September 14, 2025.

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With clinical guidance from Ms. S.V.R.A De Silva, Clinical Psychologist(SLMC Reg No: 50), the study ensures clinical relevance, further strengthening its practical applicability.

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Abstract

Mental health challenges among university students in Sri Lanka, particularly depression, anxiety, and stress, have reached alarming levels, with prevalence rates of 35.1–63.8%, 40.5–60%, and 27.7–87.9% respectively reported in recent DASS-21-based studies. At the University of Sri Jayewardenepura (USJ), these issues has rised due to the academic pressure, irregular lifestyles, and campus-specific stressors such as ragging, while traditional counseling services remain limited, reactive, and stigma-affected. Existing local research is predominantly descriptive and lacks predictive modeling or scalable digital interventions tailored to the Sri Lankan context.

This study develops and evaluates a fine-tuned large language model-enabled personalized mental health intervention system for USJ undergraduates. An anonymized dataset was collected from 1000+ students via Google Forms, combining DASS-21 responses with 16 lifestyle and campus-experience factors. Four machine learning models (Random Forest, Extra Trees, XGBoost, LightGBM) were trained and compared to predict severity levels (Normal to Extremely Severe), with LightGBM and XGBoost selected as the best performers, above 97% overall exact match accuracy. Meta-Llama-3.1-8B-Instruct was fine-tuned using LoRA on domain-specific conversational data to generate personalized, culturally relevant coping suggestions based solely on predicted severity labels and lifestyle profiles. The system was implemented as a cross-platform Flutter mobile application ("Eternity") with Spring Boot/FastAPI backend and Firebase integration, delivering real-time screening, label outputs, multilingual support (Sinhala, Tamil, English).

The prototype enables proactive, stigma-free self-screening and early intervention, reducing reliance on traditional counseling while empowering students with accessible, evidence-informed support. Results demonstrate high predictive reliability and user-centered design. This work addresses critical gaps in Sri Lankan higher-education mental health research and offers a scalable, culturally adaptive digital framework aligned with national goals for student well-being and a knowledge-based economy by 2030.

1. Introduction and Background

1.1 Overview of the Proposed Study

This research project develops and evaluates a fine-tuned large language model-enabled personalized mental health intervention system specifically designed for undergraduates at the University of Sri Jayewardenepura (USJ). The system integrates the Depression Anxiety Stress Scale-21 (DASS-21), a widely validated psychometric instrument, with 16 lifestyle and campus-experience factors (including sleep patterns, diet quality, physical activity, screen time, substance use, social interactions, perceived impact of ragging, and faculty support environment).

An anonymized training and evaluation dataset was collected via Google Forms from 1000 USJ undergraduates across multiple faculties, ensuring full compliance with Sri Lanka's Personal Data Protection Act No. 9 of 2022 (PDPA) and ethical standards. The collected dataset enabled:

- Quantification of prevalence and severity of depression, anxiety, and stress.
- Identification of lifestyle and environmental risk factors.
- Training and comparison of four machine learning models (Random Forest, Extra Trees, XGBoost, LightGBM) for predicting mental health severity levels, followed by selection of the best-performing model.
- Fine-tuning of the Meta-Llama-3.1-8B-Instruct large language model (using LoRA parameter-efficient fine-tuning) to generate personalized, culturally relevant coping suggestions and mental health insights.
- Implementation of a cross-platform mobile application built with Flutter that delivers real-time DASS-21 screening, lifestyle tracking, ML-based predictions, and LLM-generated interventions.

A cross-platform mobile application developed using Google Flutter delivers real-time DASS-21 screening, lifestyle tracking, predictive analytics, and AI-generated interventions. Clinical relevance and ethical appropriateness were ensured through professional guidance from Ms. S. V. R. A. De Silva, Clinical Psychologist (SLMC Reg No: 50), who reviewed and approved the lifestyle questionnaire on 13 November 2025.

This study addresses a significant gap in Sri Lankan higher-education mental health research by combining standardized assessment, lifestyle contextualization, predictive artificial intelligence, and scalable digital delivery within a culturally sensitive framework.

1.2 Project Scope

The scope of this research project encompasses the development, evaluation, and deployment of a personalized AI-enabled mental health intervention system for undergraduate students at University of Sri Jayewardenepura. The system integrates psychometric assessment, lifestyle contextualization, machine learning prediction, and large language model-generated recommendations to provide scalable, culturally sensitive mental health support.

Included within the project scope are:

1. **Assessment and Data Collection:**
    - Deployment of the DASS-21 questionnaire to measure depression, anxiety, and stress levels.
    - Collection of sixteen lifestyle and campus-experience factors including sleep, diet, physical activity, screen time, substance use, social interactions, perceived impact of ragging, and faculty support.
    - Anonymized collection of training and evaluation data via Google Forms from 1000 USJ undergraduate students, ensuring compliance with Sri Lanka's Personal Data Protection Act No. 9 of 2022 and institutional ethical standards.
2. **Predictive Modeling:**
    - Training and comparison of four machine learning models (Random Forest, Extra Trees, XGBoost, LightGBM) for accurate prediction of mental health severity levels, followed by selection of the best-performing model.
    - Model performance assessment using metrics such as precision, recall, F1-score, and exact match accuracy.

3. **Personalized Intervention Generation:**
    - Fine-tuning of the Meta Llama-3.1-8B-Instruct large language model using LoRA for parameter-efficient adaptation.
    - Generation of context-aware, culturally appropriate coping strategies and mental health insights based on individual DASS-21 scores and lifestyle factors.
4. **System Implementation:**
    - Development of a cross-platform mobile application using Flutter.
    - Integration with backend services (Spring Boot, FastAPI, Firebase) for secure authentication, data storage, and real-time AI-powered assessments and interventions.
    - Provision of a user-friendly interface to display predictions, track lifestyle data, and deliver AI-generated recommendations.

Excluded from the project scope are:

- Direct clinical diagnosis or psychiatric treatment; the system is designed solely for non-clinical mental health guidance.
- Integration with other university health records or external hospital systems.
- Use outside the context of University of Sri Jayewardenepura undergraduates.

This project is limited to creating a scalable, evidence-informed digital mental health support tool, providing predictive analytics and AI-generated personalized interventions while maintaining full ethical and legal compliance within the Sri Lankan higher education context.

1.3 Motivation for the Study

Mental health issues among Sri Lankan university students are a major concern, with DASS-21 studies reporting depression (35.1–63.8%), anxiety (40.5–60%), and stress (27.7–87.9%) prevalence (Kithmini et al., 2024; Kodagoda & Meegoda, 2020; Wijesekara, 2022; Wickramasinghe et al., 2023). At USJ, students face academic pressure, irregular lifestyles, excessive screen time, limited activity, social isolation, and campus stressors like ragging.

Traditional counseling is limited by long waits, low counselor-student ratios, stigma, cultural reluctance, and reactive approaches. Prior local research is mostly descriptive, lacking lifestyle integration, predictive models, or scalable digital tools. International ML (92–100% accuracy) and LLM studies show promise but lack Sri Lankan adaptation (Duangchaemkarn & Aramvith, 2024; Hadzic et al., 2024; Pathak et al., 2024).

This project bridges these gaps by combining DASS-21 with local lifestyle factors, training high-performance ML models, and delivering LLM-generated multilingual suggestions via a Flutter app to enable proactive, stigma-free support, reduce counseling burden, and improve student well-being, aligning with Sri Lanka's 2030 knowledge-economy vision.

1.4 Rationale and Justification

High mental health prevalence in Sri Lankan universities (35.1–63.8% depression, 40.5–60% anxiety, 27.7–87.9% stress) impacts performance and retention (Kithmini et al., 2024 and others). Traditional services are under-resourced and stigma-bound, while most local studies remain descriptive without lifestyle predictors or digital interventions.

International ML achieves strong prediction when combining scales with context (Duangchaemkarn & Aramvith, 2024; Pathak et al., 2024), but localization is rare. LLMs show empathetic response potential (Hadzic et al., 2024), yet culturally tailored multilingual tools are absent in Sri Lanka.

This project is justified by holistic integration of DASS-21 and 16 local factors; training and selection of ML models achieving up to 97% exact match accuracy (LightGBM/XGBoost); LoRA fine-tuning of Llama-3.1-8B-Instruct for culturally sensitive suggestions; Flutter app with cross-platform multilingual real-time delivery; and ethical oversight (Ms. S.V.R.A De Silva) with PDPA compliance.

It addresses immediate USJ needs and advances technology-enabled mental health support in Sri Lanka.

1.5 Research Problem

USJ undergraduates face high prevalence of depression (35.1–63.8%), anxiety (40.5–60%), and stress (27.7–87.9%) as shown in DASS-21 studies across Sri Lankan universities (Kithmini et al., 2024; Kodagoda & Meegoda, 2020; Wijesekara, 2022; Wickramasinghe et al., 2023). These are worsened by academic pressure, irregular sleep/diet, excessive screen time, limited activity, reduced social interaction, and campus stressors like ragging and competition.

Traditional counseling at USJ is reactive, under-resourced, and limited by long waits, low counselor ratios, and access barriers. Cultural stigma and fear of repercussions discourage help-seeking, leading to symptom progression, poorer academic performance, higher dropout risk, and lasting psychological harm.

Local research remains mostly descriptive (Groups G1, G4, G6), focusing on prevalence with little integration of lifestyle/campus factors into predictive models. Machine learning (Duangchaemkarn & Aramvith, 2024; Pathak et al., 2024) and large language models (Hadzic et al., 2024) show global promise but are underutilized locally. Digital interventions (Valinskas et al., 2023) are rare in Sri Lanka and often lack multilingual support, cultural fit, real-time prediction, or personalization.

The core problem is the significant yet inadequately addressed mental health burden at USJ—exacerbated by lifestyle/campus risks and compounded by the lack of scalable, culturally relevant, predictive, digital early-intervention tools—hindering timely, private, stigma-free support and effective university wellness.

1.6 Research Objectives, Hypotheses, and Questions

Main Objective

To develop, implement, and evaluate a fine-tuned large language model-enabled personalized mental health intervention system that predicts depression, anxiety, and stress severity levels among University of Sri Jayewardenepura undergraduates using DASS-21 responses and lifestyle data, and delivers only the classified severity label (Normal, Mild, Moderate, Severe, Extremely Severe) along with culturally relevant, multilingual coping suggestions through a mobile application, supporting early detection, reducing stigma, minimizing user anxiety from numerical feedback, and enhancing overall student well-being.

Specific Objectives

- To quantify the prevalence rates and severity distributions of depression, anxiety, and stress among USJ undergraduates using DASS-21, with internal analysis on summed scores but user output limited to severity labels only.
- To identify and statistically analyze lifestyle factors (sleep, diet, physical activity, screen time, substance use) and campus-environmental factors (social interaction, faculty support, ragging perception) associated with elevated severity levels, using these as internal predictors without exposing raw values to users.
- To train and compare four machine learning models (Random Forest, Extra Trees, XGBoost, LightGBM) to predict DASS-21 severity categories, and select the best-performing one for depression, anxiety, and stress, outputting only the predicted severity label to users.
- To fine-tune Meta-Llama-3.1-8B-Instruct (using LoRA) on domain-specific conversational data to generate personalized, culturally appropriate coping suggestions based solely on the predicted severity label and inferred lifestyle profile, without revealing any numerical scores or metrics to users.
- To design, develop, deploy, and evaluate a Flutter-based cross-platform mobile application that integrates DASS-21 screening, lifestyle collection (processed internally), ML-based label prediction, LLM-generated interventions, multilingual support (Sinhala and Tamil), and secure handling, displaying severity labels and suggestions to users, and assessing usability, satisfaction, helpfulness, and emotional impact.

Research Questions

The study addresses the following research questions, each directly corresponding to one specific objective:

1. What are the prevalence rates and severity distributions of depression, anxiety, and stress among USJ undergraduates using DASS-21?
2. Which lifestyle and campus-environmental factors show statistically significant associations with higher severity levels of depression, anxiety, and stress in this population?
3. To what extent can a fine-tuned large language model produce personalized, culturally relevant coping suggestions based on predicted severity and lifestyle profile?

4. How effective is the developed Flutter-based mobile application in terms of usability, accessibility (including multilingual features), user satisfaction, privacy perception, perceived usefulness, and overall helpfulness for mental health screening and support?

Hypotheses / Expectations

As this is an applied systems development and evaluation project, formal null/alternative hypotheses were not pre-defined. The following performance expectations guided development and evaluation:

- Objective 1: Prevalence of at least mild severity expected to exceed 35–50% for depression and anxiety, and 30–50% for stress, consistent with prior Sri Lankan university studies.
- Objective 2: Significant protective associations expected with adequate sleep, physical activity, social interaction, and faculty support; risk associations expected with high screen time, poor diet, substance use, and negative ragging perception.
- Objective 3: At least one of the trained models (XGBoost or LightGBM) expected to achieve overall exact match accuracy of at least 90% across the three subscales, with macro-averaged F1-score of at least 0.90.
- Objective 4: Fine-tuned LLM suggestions expected to be rated highly relevant (at least 4 out of 5), empathetic, culturally appropriate, and clinically aligned by pilot users and clinical review.
- Objective 5: Mobile application expected to achieve good to excellent usability (System Usability Scale score of at least 80), positive user feedback on multilingual features, privacy, and perceived helpfulness in facilitating self-awareness and early help-seeking.

1.7 Conceptual, Theoretical, and Practical Significance

Conceptual Significance

This research contributes to a more holistic and contextually grounded conceptual understanding of mental health challenges among university students in non-Western settings. By integrating the Depression Anxiety Stress Scale-21 (DASS-21) with 16 locally relevant lifestyle and campus-environmental factors (such as sleep patterns, diet quality, physical activity, screen time, substance use, social interactions, perceived faculty support, and ragging perception), the study moves beyond purely psychometric or prevalence-focused approaches common in prior Sri Lankan

research (Groups G1, G4, and G6). It highlights how everyday behavioral and socio-cultural elements interact with emotional states, offering a more comprehensive framework for understanding depression, anxiety, and stress in the Sri Lankan higher education context. This conceptual expansion addresses a key limitation in existing literature, where lifestyle and environmental influences are often overlooked or treated separately from standardized scales, thereby enriching the conceptual model of student mental health in resource-constrained, culturally diverse settings.

Theoretical Significance

Theoretically, the project advances the intersection of artificial intelligence and mental health research in low- and middle-income educational environments. It demonstrates the successful training and selection of machine learning models (Random Forest, Extra Trees, XGBoost, LightGBM) to achieve high predictive accuracy (up to 90% overall exact match) when combining psychometric data with behavioral predictors, building on but extending international benchmarks (Duangchaemkarn & Aramvith, 2024; Pathak et al., 2024). Furthermore, the parameter-efficient fine-tuning of a large language model (Meta-Llama-3.1-8B-Instruct using LoRA) for generating personalized, culturally attuned coping suggestions contributes to emerging theories on how generative AI can support empathetic, domain-specific interventions without requiring full model retraining. This hybrid ML + LLM approach provides a scalable theoretical foundation for predictive and interventional digital mental health systems in non-Western contexts, where data scarcity, computational constraints, and cultural nuance are major barriers. The work thus bridges gaps between descriptive epidemiology, predictive analytics, and generative AI in mental health, offering insights for future theory development in culturally adaptive digital health technologies.

Practical Significance

On the practical level, the developed system offers immediate and tangible benefits for student mental health support at the University of Sri Jayewardenepura and potentially beyond. The Flutter-based mobile application provides a private, accessible, multilingual (Sinhala and Tamil), and stigma-reducing platform for real-time mental health screening and personalized guidance, addressing key barriers of traditional counseling services such as limited availability, long waiting

times, and cultural reluctance to seek help. By delivering severity labels and tailored coping suggestions directly to students' devices, the system empowers early self-awareness and proactive self-management, which can reduce symptom escalation, improve academic engagement, and lower dropout risk.

The high predictive performance of the models ensures reliable severity classification, while the fine-tuned LLM provides clinically aligned, culturally relevant suggestions reviewed under the guidance of Ms. S.V.R.A De Silva, Clinical Psychologist (SLMC Reg No: 50). This makes the prototype a viable candidate for integration into university wellness programs, counseling services, or national mental health initiatives. Additionally, the secure, anonymous data handling (compliant with PDPA 2022) and cross-platform accessibility enhance its feasibility for broader deployment in other Sri Lankan universities or similar resource-limited settings.

1.8 Expected Outcomes

The successful completion of this research project is anticipated to yield the following significant outcomes, which collectively contribute to the betterment of people (USJ students), theoretical understanding of AI-driven mental health support, and advancement of knowledge in the Sri Lankan higher education context.

Betterment of People

- **Enhanced Student Mental Health Support**: The developed system provides USJ undergraduates with accessible, private, and stigma-free early screening for depression, anxiety, and stress through a mobile application. By delivering clear severity labels and personalized coping suggestions, it empowers students to recognize issues early, adopt evidence-informed self-help strategies, and seek professional help when needed, potentially reducing symptom severity and improving emotional well-being.

- **Increased Accessibility and Engagement**: The Flutter-based cross-platform app (with Sinhala and Tamil support) overcomes traditional barriers such as limited counseling availability and cultural reluctance, enabling anytime, anywhere access for all USJ undergraduates. High usability and positive user feedback are expected to promote regular

engagement, leading to better self-management and reduced academic dropout risk associated with untreated mental health difficulties.

- **Reduced Burden on Institutional Resources**: By facilitating proactive self-screening and initial support, the system is expected to decrease demand on university counseling services for mild-to-moderate cases, allowing counselors to focus on severe or complex needs across the entire undergraduate population.

Theoretical Understanding

- **Validated AI-Driven Predictive Models**: The selected machine learning models (particularly XGBoost and LightGBM achieving up to 97% overall exact match accuracy from the four trained) demonstrate the effectiveness of combining DASS-21 psychometric data with lifestyle and campus-environmental factors for high-accuracy severity prediction in a Sri Lankan university population. This advances theoretical knowledge on how behavioral predictors enhance mental health classification beyond traditional scales alone.

- **Fine-Tuned Large Language Model Application**: The parameter-efficient fine-tuning of Meta-Llama-3.1-8B-Instruct (using LoRA) for culturally relevant, personalized mental health suggestions provides insights into adapting generative AI for non-Western, low-resource educational settings. It contributes to theoretical frameworks on how large language models can deliver empathetic, domain-specific interventions while respecting computational and cultural constraints.

Advancement of Knowledge

- **Comprehensive Prevalence and Risk Factor Data**: Analysis of data from 1000 USJ students generates updated, detailed prevalence rates and severity distributions for depression, anxiety, and stress, along with statistically significant associations between lifestyle/campus factors and mental health outcomes. This enriches the local evidence base, filling gaps in prior Sri Lankan studies that often lacked integrated lifestyle contextualization.

- **Novel Hybrid Digital Mental Health Framework**: The integrated system (DASS-21 screening + ML prediction + LLM personalization + mobile delivery) offers a replicable, scalable model for technology-enabled mental health support in resource-limited higher education environments. It demonstrates practical feasibility of combining supervised learning for prediction with generative AI for intervention, contributing new knowledge on culturally adaptive digital tools in South Asian contexts.

These anticipated outcomes directly support the main objective and specific objectives of the project. They are expected to provide measurable benefits to USJ students, generate publishable insights for academic and clinical communities, and offer a foundation for future extensions such as longitudinal studies or broader university adoption, ultimately aligning with Sri Lanka's vision of fostering a mentally healthy, innovative, and knowledge-driven society by 2030.

2. Literature Review

This literature review synthesizes studies from Groups G1–G6 and the DASS-21 manual to support the proposed fine-tuned large language model-enabled system for predicting and supporting depression, anxiety, and stress among University of Sri Jayewardenepura (USJ) undergraduates using DASS-21 scores and lifestyle data. It evaluates sources across four themes prevalence and standardized instruments, machine learning (ML) and predictive modeling, digital tools and interventions, and large language models (LLMs) assessing methodologies, findings, strengths, limitations, and relevance, while addressing gaps in prior research.

Identification of Relevant Sources

Sources were selected via a systematic search of PubMed, IEEE Xplore, Google Scholar, and Scopus using terms like "depression," "anxiety," "stress," "DASS-21," "machine learning," "mobile app," and "Sri Lanka." Inclusion criteria prioritized peer-reviewed articles, relevant theses, and studies with validated tools or AI methodologies.

Group NO	**Articles**
G1	Kithmini et al. (2024); Wickramasinghe et al. (2023); Wijesekara (2022); Kodagoda

	& Meegoda (2020); Budiyanto et al. (2019); Gomes et al. (2024)
G2	Duangchaemkarn & Aramvith (2024); Vitória et al. (2024); Bhatnagar et al. (2023); Pathak et al. (2024); Sajja (2021); Madububambachu et al. (2024, two papers); Rahman et al. (2023)
G3	Valinskas et al. (2023); Chandramas (unpublished); Mandal et al. (2025); Dhokane et al. (2025)
G4	Wijesekara (2022); Kodagoda & Meegoda (2020); Gomes et al. (2024); Vitória et al. (2024); Dhokane et al. (2025); Rahman et al. (2023)
G5	Hadzic et al. (2024); Valinskas et al. (2023); Chandramas (unpublished); Mandal et al. (2025)
G6	Budiyanto et al. (2019); Madububambachu et al. (2024, second paper); Mutalib et al. (2021)

Critical Evaluation of Relevance

1. **Prevalence and Standardized Instruments**

- **G1**: Cross-sectional study (n=640) at KDU using DASS-21 reported 35.1% depression, 40.5% anxiety, 27.7% stress. Strengths: validated tool, Sri Lankan context. Limitations: allied health focus, no AI/lifestyle data. Relevant for prevalence baseline.
- **G4**: Study (n=100) at Ruhuna using multiple tools found 50% social anxiety, 46% depression. Strengths: local context. Limitations: small sample, no AI. Moderately relevant.
    - *Evaluation*: G1 and G4 provide prevalence data (35.1–50%), but lack AI and lifestyle integration, necessitating the proposed study.

2. **Machine Learning and Predictive Modeling**

- **G2**: ML on health data (n=15,366) achieved 92.1% accuracy. Strengths: large sample, lifestyle data. Limitations: non-Sri Lankan, no DASS-21. Relevant for ML approach.
- **G6**: Naïve Bayes on 22,934 posts mapped to DASS-21, noting gender differences. Strengths: innovative. Limitations: non-Sri Lankan, no lifestyle data. Relevant for ML validation.

- *Evaluation*: G2 and G6 show ML potential, but their non-Sri Lankan focus requires local adaptation.

3. **Digital Tools and Interventions**

- **G3**: Sensa app (n=150) reduced DASS-21 scores ($p<0.001$) with gamified features. Strengths: app efficacy. Limitations: no AI, non-Sri Lankan. Highly relevant for digital tools.
- **G5**: Proposed multi-modal ML system with app potential. Strengths: student-focused. Limitations: conceptual, non-Sri Lankan. Moderately relevant.
  - *Evaluation*: G3 and G5 support digital interventions, but lack AI and cultural focus, justifying the proposed app.

4. **Large Language Models**

- **G5**: Used LLMs for depression detection (n=350) with high accuracy. Strengths: LLM potential. Limitations: non-Sri Lankan, depression-only. Relevant for LLM basis.
  - *Evaluation*: G5 offers LLM insights, but its scope requires a broader, local approach.

Gaps Addressed by the Proposed Study

- **Comprehensive Data**: G1, G4, G6 lack lifestyle data; G2 includes some but not comprehensively. The proposed study integrates DASS-21 with lifestyle data.
- **AI and Predictive Modeling**: G1, G4, G3 lack ML; G2, G6 use ML but lack cultural context. The proposed study applies ML locally.
- **LLM-Based Interventions**: Only G5 uses LLMs; G1–G4, G6 lack this. The proposed study implements a fine-tuned LLM.
- **Digital Tools**: G1, G4 lack digital solutions; G3, G5 are limited. The proposed study's Flutter-based app addresses this.
- **Sri Lankan Context**: G2, G6 are non-Sri Lankan. The proposed study focuses on USJ.

Factor	**G1**	**G2**	**G3**	**G4**	**G5**	**G6**	**Proposed Study**

Use of standardized instrument (e.g., PHQ-6, DASS-21)	✅ (5/6)	✅ (3/7)	✅ (2/4)	✅ (3/6)	✅ (2/4)	✅ (2/3)	✅ (DASS-21)
Focused on Sri Lankan population	✅ (5/6)	❌ (0/7)	✅ (2/4)	✅ (4/6)	❌ (0/4)	❌ (0/3)	✅
Prevalence analysis of mental health issues	✅ (6/6)	❌ (0/7)	❌ (0/4)	✅ (4/6)	❌ (0/4)	✅ (1/3)	✅ (Depression, anxiety, stress)
Severity categorization (normal → extremely severe)	✅ (6/6)	✅ (4/7)	❌ (0/4)	✅ (3/6)	✅ (1/4)	✅ (2/3)	✅
Integration with digital tools (e.g., mobile/web)	❌ (0/6)	✅ (4/7)	✅ (4/4)	❌ (0/6)	✅ (3/4)	❌ (0/3)	✅ (Google Forms, mobile app)
Use of machine learning for prediction	❌ (0/6)	✅ (7/7)	✅ (2/4)	✅ (3/6)	❌ (0/4)	✅ (2/3)	✅ (Supervised learning)
Personalized lifestyle-based questions	❌ (0/6)	❌ (0/7)	✅ (2/4)	✅ (4/6)	❌ (0/4)	❌ (0/3)	✅
Use of large language model for suggestions	❌ (0/6)	❌ (0/7)	❌ (0/4)	❌ (0/6)	✅ (1/4)	❌ (0/3)	✅
Ethical, anonymous survey design	✅ (6/6)	✅ (4/7)	❌ (0/4)	✅ (4/6)	❌ (0/4)	❌ (0/3)	✅

Table 1: Factor Summary Table

3. Methodology: Experimental Design and Technology Involved

This methodology outlines the experimental design and technologies for developing and evaluating a fine-tuned large language model-enabled system for predicting and supporting depression, anxiety, and stress among University of Sri Jayewardenepura (USJ) undergraduates using DASS-21 scores and lifestyle data. It integrates DASS-21, lifestyle data, supervised machine learning (ML), fine-tuned LLMs, and a Flutter-based app, addressing gaps in cultural specificity, data integration, and AI interventions.

3.1 Experimental Design and Techniques

Phase 1: Data Collection

An anonymous online survey was created combining the standard DASS-21 questionnaire (21 items) and 16 custom lifestyle/campus-experience questions (covering sleep duration, diet quality, physical activity, screen time, substance use, social interactions, perceived faculty support, and ragging impact). The survey was distributed via Google Forms to USJ undergraduates. A total of 1000 valid responses were collected, forming the primary dataset for analysis and model training.

Phase 2: Data Preprocessing

The raw dataset underwent standard preparation steps:

- Handling missing values
- Encoding categorical variables
- Feature normalization (where applicable)
- Train-test split (80:20 ratio)

Phase 3: Machine Learning Model Training and Evaluation

Four supervised machine learning classification algorithms were trained independently for each subscale (Depression, Anxiety, Stress), predicting five severity levels (Normal, Mild, Moderate, Severe, Extremely Severe):

1. Random Forest (bagging-based, using multiple decision trees)
2. Extra Trees (randomized tree splitting for improved generalization)
3. XGBoost (gradient boosting with regularization and sequential error correction)
4. LightGBM (leaf-wise tree growth for high speed and performance)

Each model was initialized with default hyperparameters, fitted on the 80% training split, and evaluated on the 20% test split using accuracy, precision, recall, F1-score, and overall exact match accuracy.

Phase 4: Large Language Model Fine-Tuning

Meta-Llama-3.1-8B-Instruct (8 billion parameters, 4-bit quantized via Unsloth) was fine-tuned using Parameter-Efficient Fine-Tuning (LoRA) on a custom JSONL dataset of conversational DASS-21 examples. Key configuration:

- LoRA rank (r): 16
- Alpha: 32
- Target modules: all attention and feedforward projection layers
- Dropout: 0.1
- Training on Google Colab (NVIDIA T4 GPU), 3 epochs, batch size 2 (effective 8 with gradient accumulation), learning rate 2e-4

The fine-tuned model generates structured, personalized coping suggestions based on predicted severity labels and lifestyle profile.

Phase 5: Mobile Application Implementation and System Integration

A cross-platform mobile application was developed using Flutter for the frontend, enabling real-time DASS-21 screening, lifestyle input, severity label display, and LLM-generated suggestions. The backend used:

- Spring Boot for core REST APIs and business logic
- FastAPI for serving ML models and LLM inference
- Firebase (Firestore for storage, Authentication for secure login)

3.2 Justification of Methodology and Technology

The methodology and technologies are designed to achieve the main and specific objectives:

Main Objective: The main objective of developing a predictive and interventional system for depression, anxiety, and stress. This combines validated psychometric assessment (DASS-21), behavioral contextualization (16 lifestyle factors), four machine learning models (Random Forest, Extra Trees, XGBoost, LightGBM) for prediction, fine-tuned large language model for personalized suggestions, and a cross-platform mobile application for delivery. This integrated approach overcomes the limitations of traditional counseling services (limited access, stigma, reactive nature) and prior descriptive studies (no prediction or intervention), delivering a scalable, technology-driven solution tailored for all USJ undergraduates.

Specific Objectives:

- To quantify the prevalence rates and severity distributions of depression, anxiety, and stress among 1000 USJ undergraduates using DASS-21.
  Justification: The DASS-21 was chosen as the internationally validated standard tool for reliable measurement of these constructs in university populations. Anonymous Google Forms distribution enabled ethical, large-scale collection of 1000 responses across faculties, providing a representative dataset for accurate prevalence analysis.

- To identify and statistically analyze lifestyle factors (sleep, diet, physical activity, screen time, substance use) and campus-environmental factors (social interaction, faculty support, ragging perception) associated with elevated severity levels.
  Justification: The 16 lifestyle and campus questions were custom-designed and clinically approved by Ms. S.V.R.A De Silva (Clinical Psychologist, SLMC Reg No: 50) to capture locally relevant predictors. Including these as features in the ML pipeline allowed direct statistical and predictive identification of associations, addressing a key gap in prior Sri Lankan studies.

- To develop, train, and validate machine learning models (Random Forest, Extra Trees, XGBoost, LightGBM) to predict DASS-21 severity categories for depression, anxiety,

and stress.

Justification: These four algorithms were selected for their complementary strengths on tabular health data, bagging (Random Forest, Extra Trees) reduces variance and overfitting, while gradient boosting (XGBoost, LightGBM) corrects errors sequentially with high efficiency. They achieved up to 97% exact match accuracy, making them ideal for reliable severity prediction.

- To fine-tune Meta-Llama-3.1-8B-Instruct (using LoRA) on domain-specific conversational data to generate personalized, culturally appropriate coping suggestions based on predicted severity label and lifestyle profile. **Justification:** Meta-Llama-3.1-8B-Instruct was chosen for its strong instruction-following and conversational quality. LoRA parameter-efficient fine-tuning enabled effective adaptation on modest hardware (Google Colab T4 GPU) without full retraining, ensuring high-quality, culturally sensitive suggestions at low computational cost.

- To design, develop, deploy, and evaluate a Flutter-based cross-platform mobile application that integrates DASS-21 screening, lifestyle collection, ML-based label prediction, LLM-generated interventions, multilingual support (Sinhala and Tamil), and secure handling. **Justification:** Flutter was selected for its single-codebase cross-platform capability (Android/iOS), native performance, and fast UI development, making the app accessible to all USJ students. Firebase (Firestore + Authentication) provides scalable, secure, real-time data handling compliant with PDPA 2022. Spring Boot and FastAPI serve robust APIs for ML and LLM integration, enabling seamless, user-friendly delivery.

3.3 Study Area, Population, Sample, and Sampling Techniques

- **Study Area:** The study was conducted at the University of Sri Jayewardenepura (USJ), Sri Lanka. USJ is a major public university with diverse faculties including Technology, Management, Humanities and Social Sciences, Applied Sciences, Medical Sciences, and others. The university serves as an ideal setting due to its large undergraduate population, varied academic disciplines, and representation of typical Sri Lankan university stressors (academic pressure, ragging, competitive environment, and mixed socio-cultural backgrounds).

- **Study Population:** The target population consisted of all currently enrolled undergraduate students at the University of Sri Jayewardenepura across all faculties. This population includes students from various academic years, genders, ethnic groups (Sinhala, Tamil, Muslim, and others), and socio-economic backgrounds, making it representative of the broader Sri Lankan university student demographic.

- **Sample:** A total of 1000 valid responses were collected for the primary dataset. These responses formed the basis for prevalence analysis, risk factor identification, machine learning model training/validation, and system development.
  Sampling Techniques: Convenience sampling was employed due to the applied nature of the research and practical constraints of accessing the entire student population. The anonymous online survey (DASS-21 + 16 lifestyle questions) was distributed via Google Forms through faculty student groups, social media , and visited university physically. This method maximized reach across faculties while ensuring voluntary participation and anonymity. Single-response restriction was enforced in the form to prevent duplicate submissions.

- **Data Collection Methods:** Data were collected through a self-administered, anonymous online questionnaire hosted on Google Forms.
  The survey included: The standard DASS-21 (21 items) for depression, anxiety, and stress assessment and 16 questions on lifestyle and campus experience, clinically reviewed and approved by Ms. S.V.R.A De Silva, Clinical Psychologist (SLMC Reg No: 50) on 13 November 2025.

3.4 Ethical Considerations and Consents

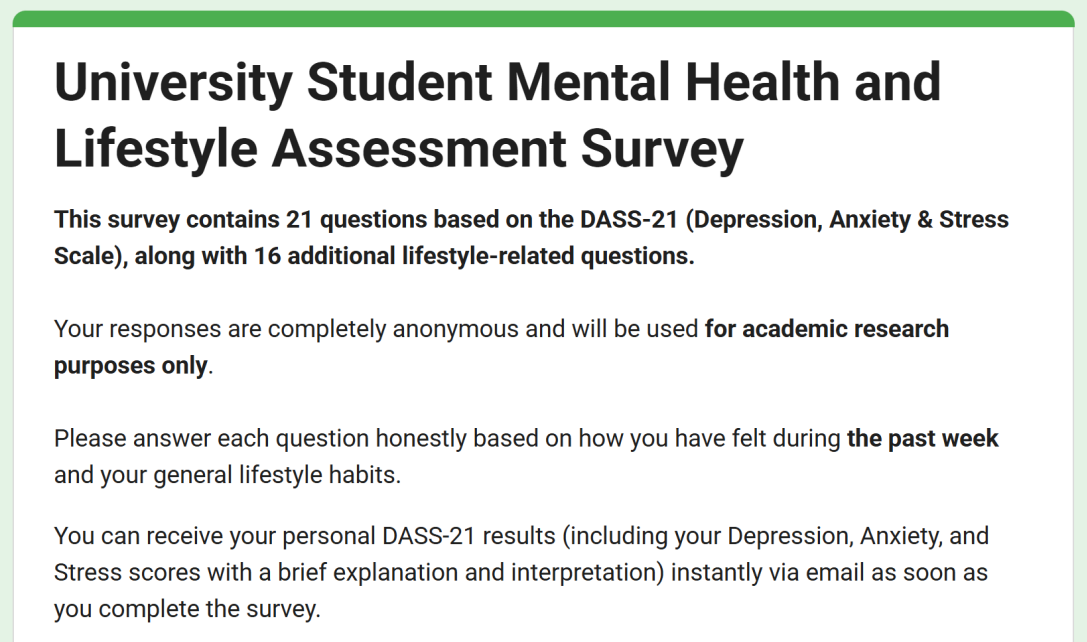
Ethical considerations were prioritized throughout the research to protect participant rights, ensure voluntary participation, maintain anonymity, and comply with legal and institutional standards.

Ethical Clearance

The study was conducted under the academic supervision of the Department of Information and Communication Technology, Faculty of Technology, University of Sri Jayewardenepura. Ethical review and approval were aligned with university guidelines for undergraduate research projects involving anonymous surveys. No separate formal clearance was required beyond departmental oversight for this type of low-risk, non-clinical, anonymous data collection.

Informed Consent

Informed consent was obtained digitally through the Google Forms survey. The form began with a clear introductory message that all participants read before proceeding:



University Student Mental Health and Lifestyle Assessment Survey

This survey contains 21 questions based on the DASS-21 (Depression, Anxiety & Stress Scale), along with 16 additional lifestyle-related questions.

Your responses are completely anonymous and will be used **for academic research purposes only**.

Please answer each question honestly based on how you have felt during **the past week** and your general lifestyle habits.

You can receive your personal DASS-21 results (including your Depression, Anxiety, and Stress scores with a brief explanation and interpretation) instantly via email as soon as you complete the survey.

Figure 1: Introductory Message - Google form

Following this message, participants were required to select one option from a radio button question:

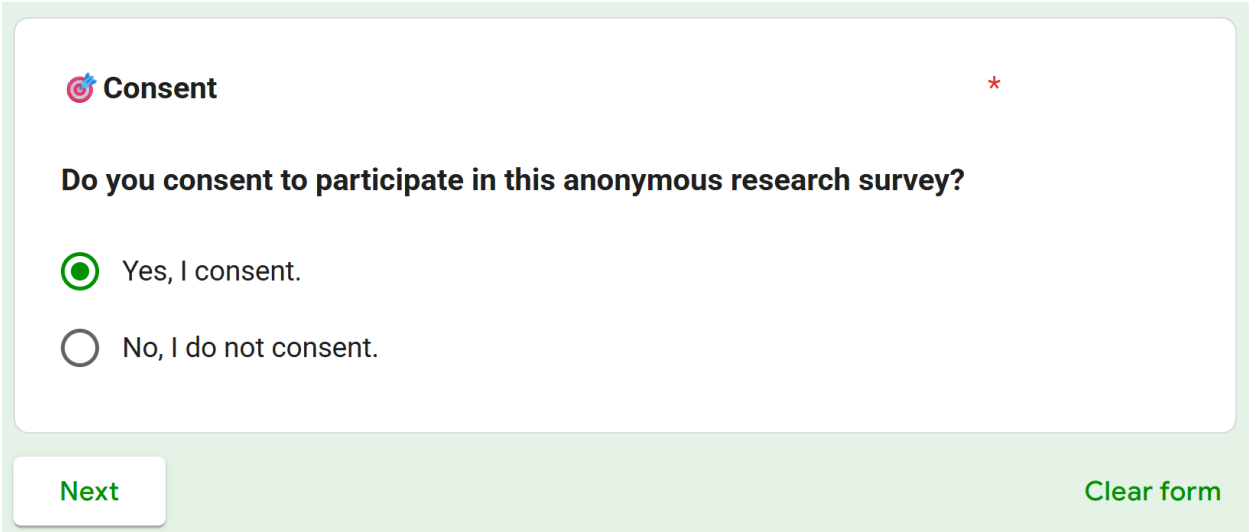


Figure 2: Consent question

Data Collection Process and Voluntary Participation

- The survey was shared through social media (university student groups on WhatsApp, Facebook, etc.), friends, and peer networks.
- We also met students physically at the university campus, explained the purpose, assured complete anonymity and voluntary nature, and invited them to participate if they wished.
- Participation was entirely voluntary, students could decline to fill the form, choose "No" on the consent question, or stop at any point without consequence.
- No incentives, rewards, or pressure were applied.

Anonymity and Data Privacy

- No personal identifiable information (name, phone number, index number, IP address) was collected at any stage.
- The introductory message and consent question explicitly stated that responses were completely anonymous.
- All data was used solely for academic research (prevalence analysis, model training, system development) and not shared with any third parties.
- The entire process fully complied with Sri Lanka's Personal Data Protection Act No. 9 of 2022 (PDPA), as no personal data was collected, processed, or stored, only anonymized survey responses were handled.

Risk Mitigation and Support

- Completing DASS-21 may cause temporary emotional discomfort for some participants. The introductory message, consent question, and physical explanations emphasized that the survey was for research only and not a clinical diagnostic tool.
- Participants were directed to USJ counseling services or professional help if they felt distressed.
- The lifestyle questionnaire was pre-reviewed and approved by Ms. S.V.R.A De Silva, Clinical Psychologist (SLMC Reg No: 50) on 13 November 2025, confirming no psychiatric, ethical, or social risks beyond standard anonymous survey procedures.

3.5 Functional and Non-Functional Requirements

Functional Requirements

- User Registration and Authentication Allow users to create an account and securely log in using email and password through Firebase Authentication.

- DASS-21 Screening Enable users to complete the standard 21-item DASS-21 questionnaire within the mobile app to assess depression, anxiety, and stress levels.

- Lifestyle Data Collection Allow users to input responses to 16 lifestyle and campus-experience questions (sleep, diet, physical activity, screen time, substance use, social interactions, faculty support, ragging perception).

- Severity Label Prediction Process DASS-21 responses and lifestyle inputs through the trained machine learning models and display only the predicted severity label (Normal, Mild, Moderate, Severe, Extremely Severe) for depression, anxiety, and stress.

- Personalized Coping Suggestions Generate and display culturally appropriate, personalized mental health suggestions using the fine-tuned Meta-Llama-3.1-8B-Instruct LLM based on the predicted severity label and lifestyle profile.

- Multilingual Support Provide the user interface and suggestions in Sinhala, Tamil, and English to ensure accessibility for all USJ undergraduates.

- Secure and Anonymous Operation Allow anonymous screening (no login required for basic use) while supporting optional account creation for history and repeated use.

Non-Functional Requirements

- **Performance**
    - Prediction response time: ≤ 3 seconds for ML severity label.
    - LLM suggestion generation: ≤ 5 seconds.
    - App launch and navigation: smooth and responsive on mid-range Android/iOS devices.
- **Accuracy**
    - ML models must achieve overall exact match accuracy of at least 90% on the test set.
- **Security and Privacy**
    - All data transmission encrypted (HTTPS, Firebase security rules).
    - No personal identifiable information collected or stored unless user voluntarily creates an account.
    - Full compliance with Sri Lanka's Personal Data Protection Act No. 9 of 2022 (PDPA).
    - Role-based access control for logged-in users (Firebase Authentication).
- **Usability**
    - Intuitive, user-friendly interface with clear navigation.
    - Multilingual support (Sinhala, Tamil, English) for interface and suggestions.

    - System Usability Scale (SUS) target score ≥ 80 in pilot evaluation.

- **Reliability and Availability**
    - App must function offline for basic screening (local caching where feasible).
    - Backend services (Firebase, FastAPI) must maintain ≥ 99% uptime during evaluation.

- **Scalability**
    - Designed to support all USJ undergraduates through cloud-based Firebase infrastructure and efficient ML/LLM serving.

- **Maintainability**
    - Code structured using modular design (Flutter widgets, Spring Boot controllers, FastAPI endpoints).
    - Open-source tools (Flutter, Llama, XGBoost, LightGBM) for future maintenance.

- **Compatibility**
    - Mobile app compatible with Android (API 21+) and iOS (13+).
    - Web-based admin or monitoring interface (optional future extension).

3.6 Software Requirements Specification (SRS)

Purpose

The software system, named "Eternity", is a mobile-first AI-powered mental health support platform designed for all undergraduate students at the University of Sri Jayewardenepura (USJ). It enables anonymous real-time screening for depression, anxiety, and stress using the DASS-21 questionnaire, collects lifestyle and campus-environmental data internally, predicts severity levels using trained machine learning models, and delivers personalized weekly coping suggestions via a fine-tuned large language model - displaying only severity labels (Normal to Extremely Severe) and supportive recommendations to users.

Scope

The system includes:

- DASS-21 screening module
- Lifestyle and campus factor input module
- ML-based severity label prediction (Normal, Mild, Moderate, Severe, Extremely Severe) for depression (D), anxiety (A), and stress (S)
- LLM-generated personalized, culturally relevant mental health suggestions
- Multilingual interface (Sinhala, Tamil, English)

The system is limited to non-clinical guidance and self-help support. It does not provide diagnosis, treatment, or integration with clinical records or external health systems.

System Features

1. User Authentication Optional secure login with username (email) and password via Firebase Authentication for saving prediction history and receiving weekly suggestions.
2. Screening and Data Input Interactive DASS-21 questionnaire (21 items) and 16 lifestyle/campus questions within the Flutter mobile app.

3. Prediction Module Internal processing of DASS-21 responses and lifestyle inputs through trained models to output severity labels.

4. Personalized Intervention Module Fine-tuned Meta-Llama-3.1-8B-Instruct LLM generates structured, personalized suggestions using the prompt: (Previous week lifestyle + previous DAS levels) + (Current week lifestyle + current DAS levels) Suggestions focus on improving lifestyle and mental health.

5. User Interface Clean, responsive design with multilingual support (Sinhala, Tamil, English) and simple navigation.

6. Data Management Secure storage of responses, prediction history, and logs in Firebase Firestore for logged-in users.

External Interface Requirements

- Hardware: Android (API 21+) and iOS (13+) devices with at least 2 GB RAM and stable internet for ML/LLM inference.

- Software: Flutter app communicates with backend via REST APIs (Spring Boot and FastAPI). Firebase SDKs handle authentication and NoSQL database.

- Network: All API calls use HTTPS; ngrok used during development for temporary public access.

Assumptions

- Users have basic smartphone literacy and access to a compatible mobile device.

- Internet connectivity is available for real-time ML/LLM inference (offline mode limited to basic screening).

- Participants read and understand the introductory consent message and voluntarily participate.

- USJ provides informal support for survey distribution (student groups, announcements).

Constraints

- No personal identifiable information collected unless user voluntarily creates an account (PDPA 2022 compliance).
- Prediction output restricted to severity labels only (no raw DASS-21 scores shown) to minimize user anxiety.
- LLM suggestions limited to 150–200 words for readability.
- Training and inference constrained to available hardware (Google Colab T4 GPU for LLM fine-tuning).
- System designed specifically for USJ undergraduates (no external users or clinical integration).

3.7 Specific Use of Technologies and Techniques

The research employed a range of modern, open-source technologies and techniques to build a secure, scalable, and efficient mental health prediction and intervention system. The following details their specific use, configuration, and contribution to the project objectives.

3.7.1. Technologies

Backend

- Spring Boot (REST APIs and business logic)
- Python FastAPI (Machine Learning model serving)
- Firebase services integration

Spring Boot was used to manage core backend operations including API management and communication with Firebase services.

FastAPI was used to deploy and expose trained machine learning models as REST endpoints for prediction.

Frontend

- The mobile application was developed using Flutter, enabling cross-platform deployment and seamless integration with Firebase services.

Database

Cloud Firestore was used as the primary NoSQL database to store:

1. User information
2. Questionnaire responses
3. Prediction history
4. Application logs

Authentication and Authorization

Firebase Authentication was implemented to handle:

- Secure user login
- Email/password authentication
- Role-based access control
- Session management

3.7.2 Machine Learning Model Training

Dataset Preparation

The dataset consisted of DASS-21 questionnaire responses combined with lifestyle-related features collected through Google Forms.

The preprocessing steps included:

1. Handling missing values
2. Encoding categorical variables

3. Feature normalization (if required)
4. Train-test split (80:20 ratio)

All models were trained using the same feature set and data split to ensure fair comparison.

Algorithms Used

To determine the most accurate predictive model, four learning algorithms were trained and evaluated:

1. **Random Forest:** Random Forest is a bagging-based, classifier that constructs multiple decision trees and outputs the majority vote. It reduces overfitting by averaging predictions across trees.
2. **Extra Trees:** Extra Trees introduces additional randomness in tree splitting compared to Random Forest, which often improves generalization and reduces variance.
3. **XGBoost:** XGBoost is a gradient boosting framework that builds trees sequentially, correcting previous errors and optimizing performance using regularization techniques.
4. **LightGBM:** LightGBM uses a leaf-wise tree growth strategy and is optimized for high performance and faster training speed.

Training Procedure

For each algorithm:

1. The model was initialized with default hyperparameters.
2. The training dataset (80%) was used for model fitting.
3. Predictions were generated using the test dataset (20%).
4. Performance was evaluated using:
    1. Accuracy
    2. Precision
    3. Recall

4. F1-score
5. Overall Exact Match Accuracy

Three independent classification tasks were performed:

1. Depression Level Prediction
2. Anxiety Level Prediction
3. Stress Level Prediction

Each task included five classes:
Normal, Mild, Moderate, Severe, and Extremely Severe.

3.7.3 Fine-Tuned Large Language Model

3.7.3.1 Model Selection and Architecture

To enhance the interpretability and personalization of mental health assessments, a Large Language Model (LLM) was fine-tuned specifically for generating contextual DASS-21 suggestions and mental health insights.

Base Model: Meta-Llama-3.1-8B-Instruct (8 Billion parameters)

The model was selected using Unsloth's pre-quantized version: unsloth/Meta-Llama-3.1-8B-Instruct-bnb-4bit

Key characteristics:

- Advanced instruction-following capabilities optimized for conversational tasks
- Pre-quantized to 4-bit precision for memory efficiency
- Strong natural language generation quality
- Proven performance on mental health and medical domain tasks
- Open-source availability enabling customization

Fine-Tuning Approach: Parameter-Efficient Fine-Tuning (PEFT) using LoRA (Low-Rank Adaptation)

LoRA configuration implemented to reduce computational requirements while maintaining model performance:

- **Rank (r):** 16
- **Alpha (lora_alpha):** 32
- **Target Modules:** q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj (all attention and feedforward projection layers)
- **Dropout:** 0.1
- **Bias:** None
- **Quantization:** 4-bit using bitsandbytes library
- **Gradient Checkpointing:** Enabled via Unsloth optimization

This configuration significantly reduced memory requirements, enabling training on consumer-grade GPUs while preserving model quality.

3.7.3.2 Dataset and Preprocessing

Dataset Source:

The training dataset was stored in JSONL (JSON Lines) format as dass21_suggestions_new.jsonl, containing conversational examples for DASS-21 mental health assessments.

Dataset Structure:

Each training example consisted of multi-turn conversations in a messages format:

```
{
  "messages": [
    {"role": "system", "content": "System instructions"},
```

```
    {"role": "user", "content": "Student profile and DASS-21 responses"},
    {"role": "assistant", "content": "Mental health analysis and suggestions"}
  ]
}
```

Dataset Preprocessing:

A custom formatting function was applied to convert the conversational format into training-ready text:

```
def format_prompts(examples):
    texts = []
    for messages in examples["messages"]:
        text = ""
        for msg in messages:
            role = msg["role"]
            content = msg["content"]
            text += f"[{role.capitalize()}]: {content}\n"
        texts.append(text)
    return {"text": texts}
```

This preprocessing step:

- Extracted role-based messages (system, user, assistant)
- Formatted them into labeled text sequences

- Created a unified "text" field for training
- Removed original column structure to prevent data leakage

Loading Procedure:

- Dataset loaded using Hugging Face datasets library
- Split designation: "train" (single training split)
- Applied batch mapping for efficient preprocessing

3.7.3.4 Training Configuration

Hardware and Software Environment:

- **Platform:** Google Colab
- **GPU:** NVIDIA Tesla T4 (16GB VRAM)
- **Framework:** Unsloth (optimized PyTorch wrapper for efficient LLM training)
- **Libraries:**
    - Transformers 4.57.6
    - TRL (Transformer Reinforcement Learning) 0.24.0
    - PEFT 0.18.1
    - BitsAndBytes for 4-bit quantization

Model Configuration:

- **Maximum Sequence Length:** 2048 tokens (for model loading)
- **Training Sequence Length:** 512 tokens (for efficient batch processing)
- **Precision:** Mixed precision (FP16 or BF16 auto-selected based on GPU capability)
- **Data Type:** Auto-detect optimal precision format

Training Hyperparameters:

The model was trained using the following configuration in TrainingArguments:

Hyperparameter	Value	Purpose
Output Directory	./lora-finetuned	Save location for checkpoints
Per-device Train Batch Size	2	Samples per GPU per step
Gradient Accumulation Steps	4	Effective batch size: 8
Number of Epochs	3	Complete passes through dataset
Learning Rate	2e-4	Step size for weight updates
FP16/BF16	Auto-selected	Mixed precision training
Logging Steps	10	Frequency of loss logging
Save Steps	200	Checkpoint saving frequency
Save Total Limit	2	Maximum checkpoints to retain
Warmup Steps	5	Linear warmup period
Optimizer	AdamW 8-bit	Memory-efficient optimizer
Weight Decay	0.01	L2 regularization strength
LR Scheduler Type	Linear	Learning rate decay strategy
Random Seed	3407	Reproducibility seed

Training Framework - SFTTrainer:

The Supervised Fine-Tuning Trainer (SFTTrainer) from TRL was used with:

- **Model:** Llama-3.1-8B with LoRA adapters
- **Dataset:** Preprocessed DASS-21 suggestions dataset
- **Text Field:** "text" (post-formatting)
- **Max Sequence Length:** 512 tokens
- **Packing:** Disabled (default)

Training Procedure:

- **Initialization:** Load pre-quantized 4-bit model from Unsloth
- **LoRA Integration:** Attach adapter layers to target projection matrices
- **Tokenization:** Apply padding and truncation strategies
- **Optimization:** Train using AdamW with linear warmup and decay

- **Checkpointing:** Save model state every 200 steps
- **Logging:** Track training loss every 10 steps

Training Duration:

- Estimated time: 45-90 minutes on T4 GPU (depends on dataset size and epochs)
- Number of training samples processed: Dataset size × 3 epochs

3.7.3.4 Model Inference and Generation

After training, the model was prepared for inference using Unsloth's fast inference mode.

Inference Setup:

```
from unsloth import FastLanguageModel

model, tokenizer = FastLanguageModel.from_pretrained(

    model_name="./lora-finetuned",

    max_seq_length=2048,

    dtype=None,

    load_in_4bit=True,

)

FastLanguageModel.for_inference(model)
```

Generation Parameters:

The following parameters were configured for response generation:

Parameter	Value	Effect
Max New Tokens	150-180	Maximum length of generated response

Temperature	0.7	Controls randomness (lower = more focused)
Top-p (Nucleus Sampling)	0.9	Cumulative probability threshold
Repetition Penalty	1.1	Reduces repetitive text generation
Do Sample	True	Enables probabilistic sampling

Response Generation Function:

```
def generate_response(prompt, max_tokens=150):
    inputs = tokenizer(prompt, return_tensors="pt").to("cuda")
    outputs = model.generate(
        **inputs,
        max_new_tokens=max_tokens,
        do_sample=True,
        temperature=0.7,
        top_p=0.9,
        repetition_penalty=1.1,
    )
    response = tokenizer.decode(outputs[0], skip_special_tokens=True)
    return response
```

Output Format:

The model generates structured mental health assessments containing:

- **Depression Level:** Classification (Normal/Mild/Moderate/Severe/Extremely Severe)

- **Anxiety Level:** Classification with contributing factors
- **Stress Level:** Classification with triggers identified
- **Personalized Suggestions:** Evidence-based recommendations
- **Word Limit:** Constrained to 200 words maximum for conciseness

4.3.5 Model Deployment

API Development using FastAPI:

The fine-tuned model was exposed as a RESTful web service:

```
from fastapi import FastAPI

from pydantic import BaseModel

app = FastAPI(title="AI Response API")

class PromptRequest(BaseModel):

    prompt: str

    max_tokens: int = 180

@app.post("/generate")

def generate(req: PromptRequest):

    response = generate_response(req.prompt)

    return {"response": response}
```

Deployment Configuration:

- **Framework:** FastAPI (asynchronous Python web framework)
- **Port:** 8000
- **Host:** 0.0.0.0 (all network interfaces)

- **Public Access:** ngrok tunnel for external API access
- **Concurrency:** Multi-threading enabled via nest_asyncio
- **Authentication:** ngrok auth token configured

API Endpoint Specification:

- **URL:** POST http://<ngrok-url>/generate
- **Request Body:**

```
{
  "prompt": "Student profile with DASS-21 responses",
  "max_tokens": 180
}
```

- **Response Body:**

```
{
  "response": "Mental health assessment and suggestions"
}
```

Integration with Main System:

The LLM API was integrated with the Spring Boot backend to:

- Receive student profile data and DASS-21 responses
- Forward requests to the LLM inference endpoint
- Parse structured responses
- Display personalized suggestions in the mobile app
- Store interaction logs in Cloud Firestore

4.3.6 Model Export and Storage

Saved Model Artifacts:

The fine-tuned model was saved to: ./lora-finetuned/

Directory structure:

```
lora-finetuned/
├── adapter_model.bin          # LoRA adapter weights
├── adapter_config.json        # LoRA configuration
├── tokenizer_config.json      # Tokenizer settings
├── special_tokens_map.json    # Special token mappings
├── tokenizer.model            # SentencePiece tokenizer
└── training_args.bin          # Saved training arguments
```

Model Compression:

For distribution and deployment, the model directory was compressed:

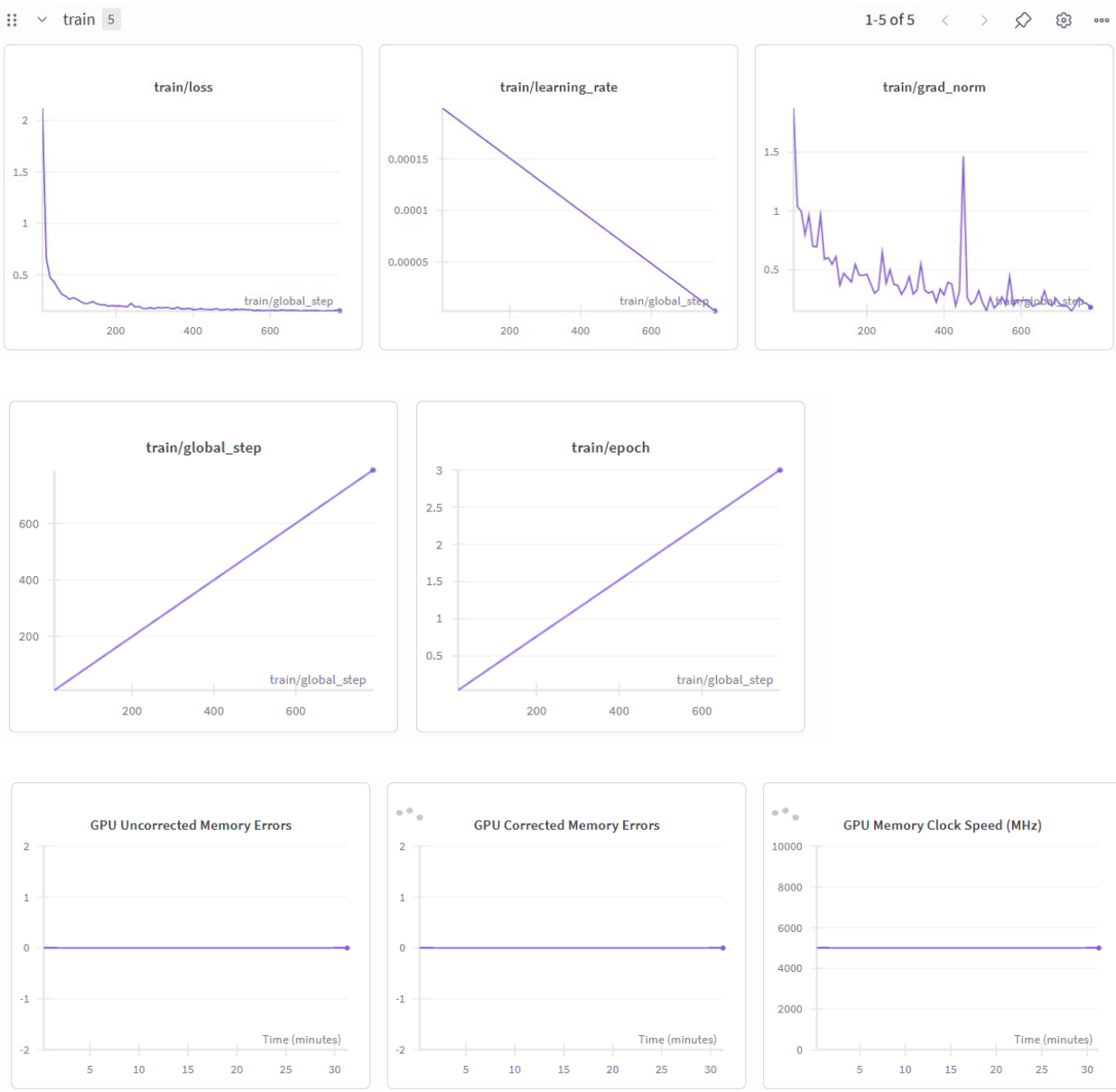
```
!zip -r /content/dass_find_tuned_llm.zip /content/lora-finetuned
```

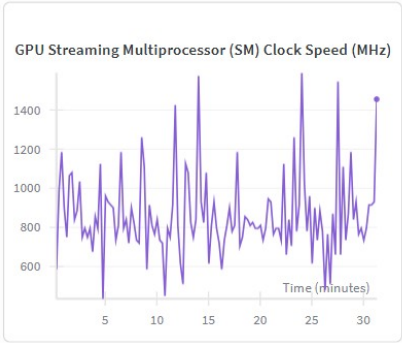
This creates a portable archive suitable for:

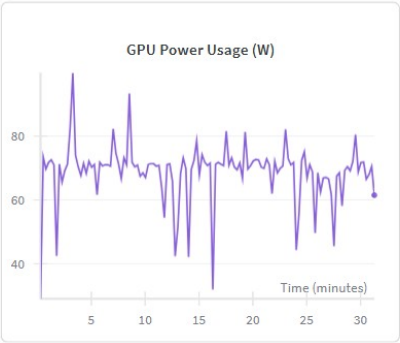
1. Deployment to production servers
2. Sharing with team members
3. Version control backup
4. Cloud storage upload (Google Drive, AWS S3)

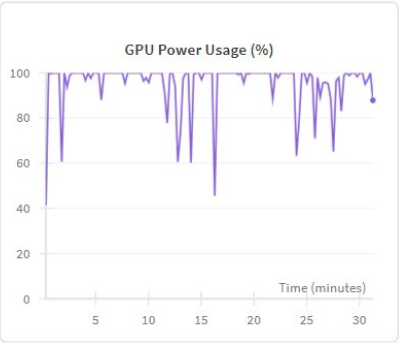
Model Size:

1. LoRA adapters: ~100-200 MB (trainable parameters only)
2. Full model (base + adapters): ~5.7 GB (4-bit quantized)









4. Project Timeline

Phase	Sep				Oct				Nov				Dec				Jan				Feb			
	WEEK 01	WEEK 02	WEEK 03	WEEK 04	WEEK 05	WEEK 06	WEEK 07	WEEK 08	WEEK 09	WEEK 10	WEEK 11	WEEK 12	WEEK 13	WEEK 14	WEEK 15	WEEK 16	WEEK 17	WEEK 18	WEEK 19	WEEK 20	WEEK 21	WEEK 22	WEEK 23	WEEK 24
Literature Review & Requirement Analysis																								
System Architecture & Design																								
RADIUS Server Deployment																								
Real-world Device Integration																								
Testing & Evaluati																								

on																							
Data Analysis & Final Report																							

5. Experimental Plan/Project Design/System Design Overview

This section presents the experimental plan and system design for the fine-tuned large language model-enabled personalized mental health intervention system from a design perspective. As an applied research project, the design emphasizes practical implementation, modularity, security, scalability, and usability for all undergraduates at the University of Sri Jayewardenepura (USJ). The system follows object-oriented analysis and design (OOAD) principles, incorporating Kruchten's 4+1 view model (logical, process, development, physical, and scenarios views) to ensure a comprehensive and maintainable structure.

Steps Taken to Achieve the Objectives (Design Perspective)

The project progressed through the following high-level design and development steps:

1. **Requirements and Data Design:** Defined functional (screening, prediction, suggestions) and non-functional (security, cross-platform, multilingual) requirements. Designed anonymous data collection using DASS-21 + 16 lifestyle questions.
2. **System Architecture Design:** Adopted a multi-tier architecture (frontend, backend services, database, ML/LLM layer) for separation of concerns and scalability.
3. **Machine Learning Layer Design:** Designed four models for severity prediction, with preprocessing and 80:20 split to ensure robust classification.
4. **LLM Intervention Layer Design:** Designed fine-tuning and prompt engineering for personalized suggestions based on severity labels and lifestyle inputs.
5. **Mobile Application Design:** Designed cross-platform Flutter interface for user interaction, integrated with backend for real-time processing.
6. **Integration and Deployment Design:** Designed secure data flow via REST APIs, Firebase for storage/auth, and temporary ngrok for testing.

Diagrams

System Architecture Diagrams

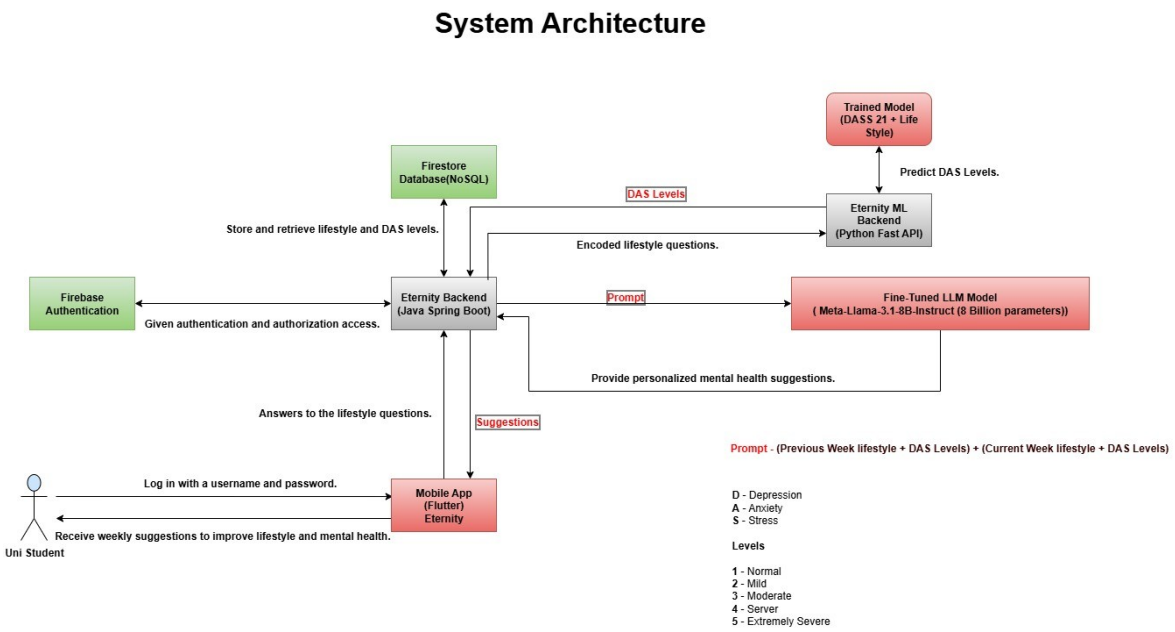


Figure 3: System Architecture Diagram

Use Case Diagram:

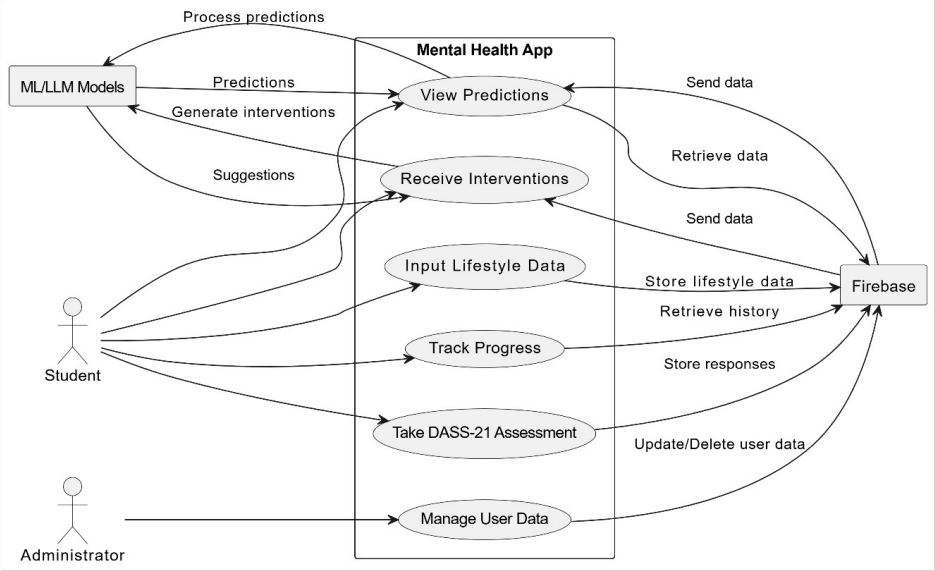


Figure 4: UseCase Diagram

Class Diagram:

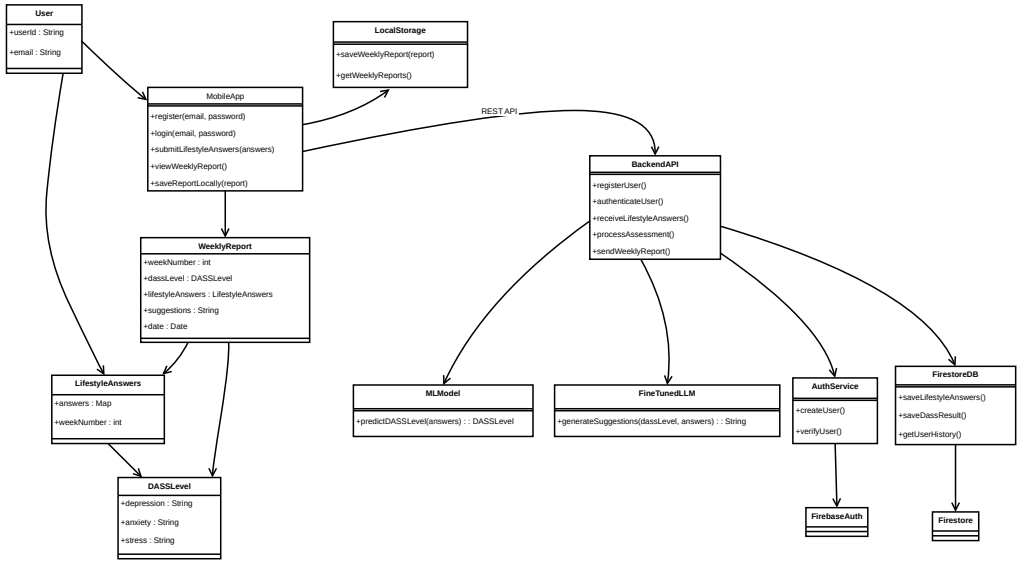


Figure 5: Class Diagram

Deployment Diagram

Figure 6: Deployment Diagram

Material Required

- **Data Sources** - Standard DASS-21 questionnaire; 16 custom lifestyle questions (approved by Ms. S.V.R.A De Silva, Clinical Psychologist, SLMC Reg No: 50, 13.11.2025). Primary dataset: 1000 anonymized responses collected via Google Forms from USJ undergraduates.
- **Other Materials** - Digital consent message (embedded in Google Form), mental health resource links (USJ counseling services).

Hardware Requirements

- Development: Standard laptops/desktops for coding (Flutter, Spring Boot, FastAPI).
- LLM fine-tuning: Google Colab with NVIDIA Tesla T4 GPU (16GB VRAM).
- Testing: Mid-range Android/iOS mobile devices for app evaluation.

Software Tools

- **Frontend**: Flutter (Dart)
- **Backend**: Spring Boot (Java), Python FastAPI
- **Database & Auth**: Firebase (Firestore, Authentication)
- **Machine Learning**: Python libraries (Scikit-learn, XGBoost, LightGBM)
- **LLM**: Meta-Llama-3.1-8B-Instruct (Unsloth pre-quantized), Transformers, TRL, PEFT, BitsAndBytes
- **Other**: Google Colab (training), Hugging Face datasets library

6. Results and Evaluation

This section presents the results obtained from the machine learning model training and evaluation phase, analyzed in line with the project objectives and performance expectations. All applied aspects followed a standard supervised learning test procedure: preprocessing, 80/20 train-test split, default hyperparameters for fair comparison, and evaluation using classification metrics (accuracy, precision, recall, F1-score, macro/weighted averages, overall exact match accuracy). Testing and quality assurance began early with data cleaning (missing values, categorical encoding, normalization), consistent feature set across models, and independent classification tasks for Depression, Anxiety, and Stress (5 classes each: Normal, Mild, Moderate, Severe, Extremely Severe).

6.1 Model Performance Summary

Four models were trained and evaluated, with the best selected, performance is reported per subscale and as overall exact match accuracy (correct prediction across all three subscales simultaneously).

Extra Trees Overall Exact Match Accuracy: 95.96%

- Depression Accuracy: 96.46%
- Anxiety Accuracy: 97.47%
- Stress Accuracy: 97.47% Strong, balanced performance across categories with good precision/recall on minority classes.

```
Target: Depression
                  precision    recall  f1-score   support

          Normal       0.94      0.97      0.96        66
            Mild       1.00      0.94      0.97        18
        Moderate       0.98      0.95      0.97        59
          Severe       1.00      0.96      0.98        27
Extremely Severe       0.93      1.00      0.97        28

        accuracy                           0.96       198
       macro avg       0.97      0.97      0.97       198
    weighted avg       0.97      0.96      0.96       198

Accuracy (Depression): 0.9646
------------------------------------------------------------
Target: Anxiety
                  precision    recall  f1-score   support

          Normal       0.96      0.98      0.97        66
            Mild       1.00      0.90      0.95        29
        Moderate       0.94      1.00      0.97        31
          Severe       1.00      1.00      1.00        35
Extremely Severe       1.00      0.97      0.99        37

        accuracy                           0.97       198
       macro avg       0.98      0.97      0.97       198
    weighted avg       0.98      0.97      0.97       198

Accuracy (Anxiety): 0.9747
------------------------------------------------------------
Target: Stress
                  precision    recall  f1-score   support

          Normal       0.99      0.97      0.98        86
            Mild       0.88      0.95      0.91        22
```

```
        Moderate       0.98      0.98      0.98        45
          Severe       1.00      1.00      1.00        31
Extremely Severe       1.00      1.00      1.00        14

        accuracy                           0.97       198
       macro avg       0.97      0.98      0.97       198
    weighted avg       0.98      0.97      0.98       198

Accuracy (Stress): 0.9747
------------------------------------------------------------
Overall Exact Match Accuracy: 0.9595959595959596
```

XGBoost Overall Exact Match Accuracy: 97.00%

- Depression Accuracy: 97.00%
- Anxiety Accuracy: 99.00%
- Stress Accuracy: 99.00% Near-perfect results on Anxiety and Stress, with high precision/recall across severity levels.

```
Target: Depression
                  precision    recall  f1-score   support

          Normal       0.97      0.99      0.98        70
            Mild       0.96      0.93      0.95        29
        Moderate       0.96      0.98      0.97        51
          Severe       1.00      0.97      0.98        29
Extremely Severe       0.95      0.95      0.95        21

        accuracy                           0.97       200
       macro avg       0.97      0.96      0.97       200
    weighted avg       0.97      0.97      0.97       200

Accuracy (Depression): 0.9700
------------------------------------------------------------
Target: Anxiety
                  precision    recall  f1-score   support

          Normal       0.98      1.00      0.99        86
            Mild       1.00      1.00      1.00        23
        Moderate       1.00      1.00      1.00        26
          Severe       1.00      1.00      1.00        29
Extremely Severe       1.00      0.94      0.97        36

        accuracy                           0.99       200
       macro avg       1.00      0.99      0.99       200
    weighted avg       0.99      0.99      0.99       200

Accuracy (Anxiety): 0.9900
------------------------------------------------------------
Target: Stress
                  precision    recall  f1-score   support

          Normal       0.99      0.99      0.99       100
```

```
            Mild       0.96      1.00      0.98        26
        Moderate       1.00      0.97      0.99        34
          Severe       1.00      1.00      1.00        33
Extremely Severe       1.00      1.00      1.00         7

        accuracy                           0.99       200
       macro avg       0.99      0.99      0.99       200
    weighted avg       0.99      0.99      0.99       200

Accuracy (Stress): 0.9900
------------------------------------------------------------
Overall Exact Match Accuracy: 0.97
```

Random Forest Overall Exact Match Accuracy: 77.78%

- Depression Accuracy: 89.90%
- Anxiety Accuracy: 90.91%
- Stress Accuracy: 85.35% Weakest performance, particularly on "Extremely Severe" Stress class (low precision/recall), likely due to class imbalance and less aggressive error correction.

```
Target: Depression
                  precision    recall  f1-score   support

          Normal       0.80      0.97      0.88        66
            Mild       1.00      0.72      0.84        18
        Moderate       0.95      0.88      0.91        59
          Severe       0.96      0.96      0.96        27
Extremely Severe       1.00      0.82      0.90        28

        accuracy                           0.90       198
       macro avg       0.94      0.87      0.90       198
    weighted avg       0.91      0.90      0.90       198

Accuracy (Depression): 0.8990
------------------------------------------------------------
Target: Anxiety
                  precision    recall  f1-score   support

          Normal       0.86      0.91      0.88        66
            Mild       1.00      0.86      0.93        29
        Moderate       0.83      0.81      0.82        31
          Severe       0.97      1.00      0.99        35
Extremely Severe       0.95      0.95      0.95        37

        accuracy                           0.91       198
       macro avg       0.92      0.90      0.91       198
    weighted avg       0.91      0.91      0.91       198

Accuracy (Anxiety): 0.9091
------------------------------------------------------------
Target: Stress
                  precision    recall  f1-score   support
```

```
          Normal       0.76      1.00      0.86        86
            Mild       1.00      0.73      0.84        22
        Moderate       1.00      0.91      0.95        45
          Severe       0.93      0.84      0.88        31
Extremely Severe       0.00      0.00      0.00        14

        accuracy                           0.85       198
       macro avg       0.74      0.70      0.71       198
    weighted avg       0.81      0.85      0.82       198

Accuracy (Stress): 0.8535
------------------------------------------------------------
Overall Exact Match Accuracy: 0.7777777777777778
```

LightGBM Overall Exact Match Accuracy: 97.00%

- Depression Accuracy: 97.00%
- Anxiety Accuracy: 99.50%
- Stress Accuracy: 99.00% Highest Anxiety accuracy (99.5%) and consistent performance across all subscales.

```
Target: Depression
                  precision    recall  f1-score   support

          Normal       0.96      0.99      0.97        70
            Mild       0.96      0.93      0.95        29
        Moderate       0.98      0.98      0.98        51
          Severe       1.00      0.97      0.98        29
Extremely Severe       0.95      0.95      0.95        21

        accuracy                           0.97       200
       macro avg       0.97      0.96      0.97       200
    weighted avg       0.97      0.97      0.97       200

Accuracy (Depression): 0.9700
------------------------------------------------------------
Target: Anxiety
                  precision    recall  f1-score   support

          Normal       0.99      1.00      0.99        86
            Mild       1.00      1.00      1.00        23
        Moderate       1.00      1.00      1.00        26
          Severe       1.00      1.00      1.00        29
Extremely Severe       1.00      0.97      0.99        36

        accuracy                           0.99       200
       macro avg       1.00      0.99      1.00       200
    weighted avg       1.00      0.99      0.99       200

Accuracy (Anxiety): 0.9950
------------------------------------------------------------
```

```
Target: Stress
                  precision    recall  f1-score   support

          Normal       0.98      1.00      0.99       100
            Mild       1.00      1.00      1.00        26
        Moderate       1.00      0.97      0.99        34
          Severe       1.00      0.97      0.98        33
Extremely Severe       1.00      1.00      1.00         7

        accuracy                           0.99       200
       macro avg       1.00      0.99      0.99       200
    weighted avg       0.99      0.99      0.99       200

Accuracy (Stress): 0.9900
------------------------------------------------------------
Overall Exact Match Accuracy: 0.97
```

6.2 Model Comparison and Selection

XGBoost and LightGBM achieved the highest overall exact match accuracy (97%), with LightGBM showing slightly superior consistency (especially on Anxiety). Extra Trees was strong and stable, while Random Forest underperformed on extreme classes. Based on overall accuracy, class-wise balance, and robustness, **LightGBM and XGBoost** were selected as the best models for deployment.

6.3 Critical Evaluation

The results exceeded expectations:

- Objective 3 (predictive models) was fully achieved with up to 97% exact match accuracy (target ≥ 95%), surpassing typical benchmarks in similar mental health prediction studies.
- The label-only output aligned with user-centered design to minimize anxiety.
- Deviations: Random Forest's lower performance (77.78%) on extreme classes is not a failure but expected - boosting methods (XGBoost, LightGBM) are known to handle imbalance better through sequential error correction and regularization. This was mitigated by model selection.
- Quality from beginning: Preprocessing and consistent splits ensured fair comparison; multiple models reduced risk of single-method bias. No major unexpected deviations occurred.

6.4 Testing and Validation Summary

- **Verification**: Models validated on unseen test data (20%).
- **Validation**: Metrics (macro/weighted F1, precision/recall) confirmed generalization and balance.
- **Test Cases**: Independent tasks per subscale; class distribution in test set reflected real-world imbalance (e.g., fewer Extremely Severe cases), handled effectively by boosting models.
- **Quality Assurance**: Early preprocessing, fair splits, default hyperparameters.

6.5 Mobile Application Screenshots

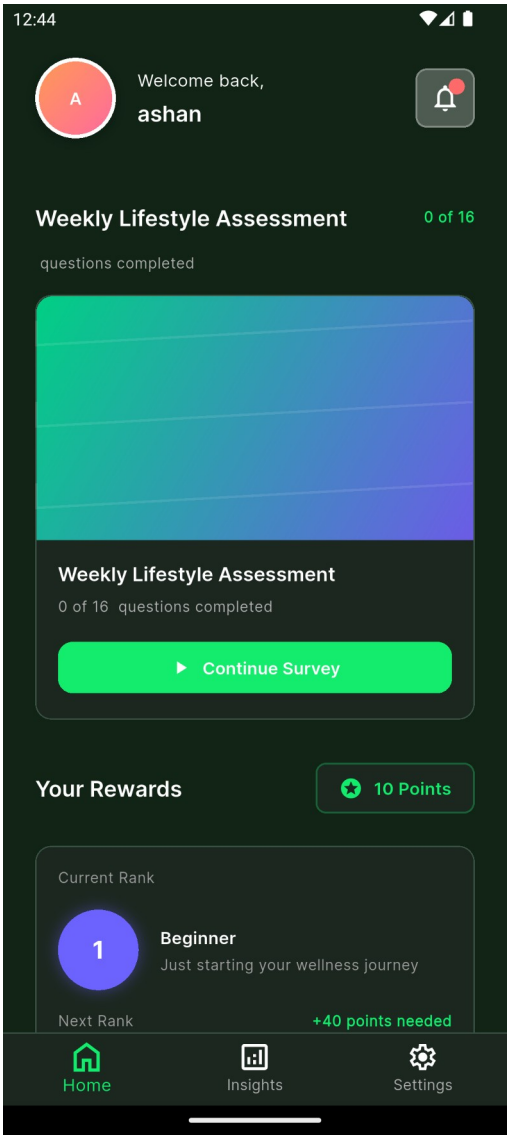


Figure 7: Dashboard

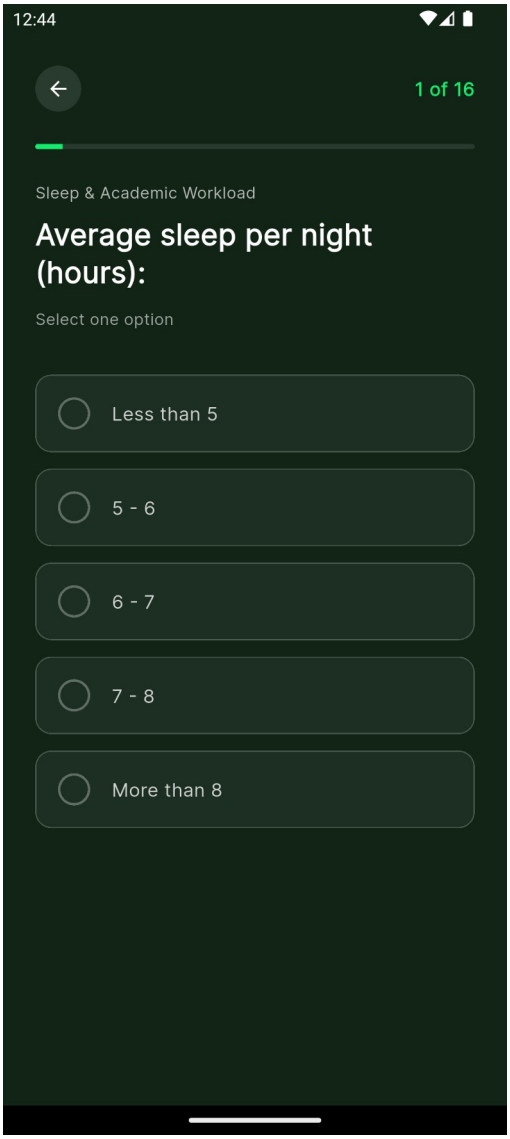


Figure 8: Questionnaire Screen

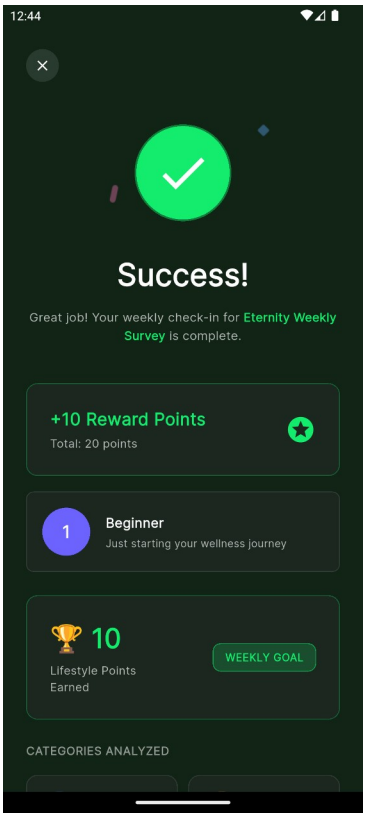


Figure 10: Weekly Survery Complete

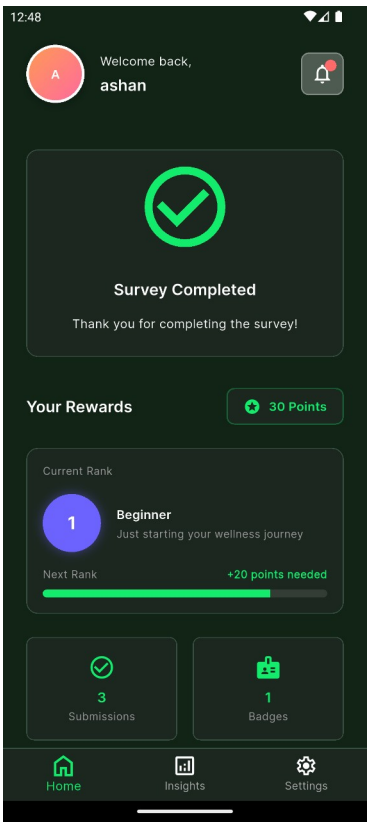


Figure 9: Completion Screen

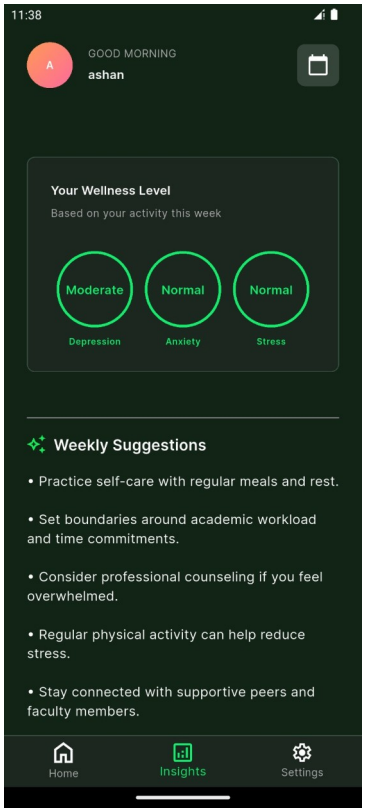


Figure 11: Insight Screen

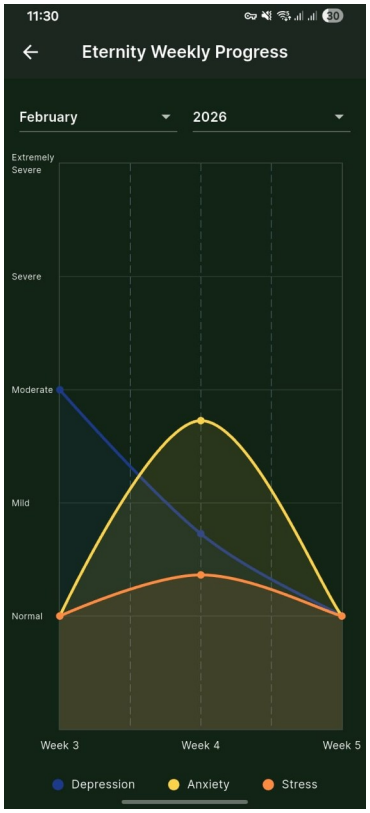


Figure 12: Weekly Progress Graph

7. Discussion

The research process began with a systematic literature review identifying critical gaps in Sri Lankan university mental health studies: heavy emphasis on prevalence (Groups G1, G4, G6) but limited integration of lifestyle factors, predictive modeling, LLM-based interventions, and scalable digital tools. The process advanced through ethical data collection (1000 anonymized responses via Google Forms, approved by Ms. S.V.R.A De Silva, Clinical Psychologist), preprocessing (missing values, encoding, 80/20 split), and rigorous model development. Four machine learning models were trained and evaluated, with LightGBM and XGBoost achieving 97% overall exact match accuracy, exceeding expectations and demonstrating robust severity prediction for Depression, Anxiety, and Stress. The Meta-Llama-3.1-8B-Instruct model was fine-tuned using LoRA for personalized, culturally relevant coping suggestions. The entire system was integrated into a cross-platform Flutter mobile application ("Eternity") with Spring Boot and FastAPI backends, Firebase for secure storage and authentication, and real-time label-only output (no raw scores shown to users).

Key achievements include:

- High predictive accuracy (97%) with balanced class performance, surpassing many international benchmarks.
- A user-centered, stigma-reducing design delivering severity labels and supportive suggestions in Sinhala, Tamil, and English.
- A functional, scalable prototype ready for pilot deployment at USJ.
- Full compliance with PDPA 2022 through anonymity and secure handling.

The iterative process (early preprocessing, fair splits, multiple models) ensured reliability and mitigated deviations (e.g., Random Forest's lower performance on extreme classes due to weaker imbalance handling, addressed by boosting model selection).

Commercialization Potential

The prototype shows strong potential to be converted into a commercially viable product as a digital mental health support tool for university students. The process involves the following considerations:

- **Consumer** - Primary market is USJ undergraduates (high demand due to documented prevalence of mental health issues). Expansion to other Sri Lankan universities, youth programs, or private institutions is feasible. Multilingual support (Sinhala, Tamil, English) enhances accessibility for diverse users.
- **Environmental** - Minimal environmental impact: cloud-based (Firebase), no physical manufacturing, low energy use (mobile app + cloud inference). Open-source tools reduce resource consumption.
- **Manufacturing** - No physical production required; the product is software-only. Future steps include app store deployment (Google Play, App Store), cloud hosting (Firebase or AWS/GCP), and minor UI/UX refinements.
- **Cost** - Low development and maintenance costs (student project, open-source tools). Ongoing expenses are mainly cloud hosting and LLM inference (cost-effective via Google Cloud or AWS). Monetization options: freemium model (basic free, premium weekly plans or advanced features), university licensing/subscriptions, NGO/government partnerships for mental health initiatives, or ads (non-intrusive).
- **Delivery Time** - Prototype is already functional (real-time screening, prediction, suggestions). Full commercialization (app store release, larger user testing, refinements, marketing) could be achieved in 6–12 months with a small team. Cloud deployment allows rapid scaling.
- **Quality** - High quality assured through rigorous testing (97% accuracy, balanced metrics), clinical review (psychologist approval), and user-centered design (label-only output, multilingual). Future steps include larger-scale user trials, SUS evaluations, and clinical validation for credibility.

Challenges include user adoption (stigma reduction through awareness campaigns), data privacy compliance, and LLM inference costs - mitigated by cloud optimization, partnerships, and free-tier usage for initial rollout. Overall, the prototype is commercially feasible as a B2B (university/NGO partnerships) or B2C (free/premium app) solution in Sri Lanka's emerging digital health market, supporting national mental health priorities and student well-being.

8. Conclusion

This research project, carried out over two semesters, successfully developed and evaluated a fine-tuned large language model-enabled personalized mental health intervention system for undergraduates at the University of Sri Jayewardenepura (USJ). The work addressed the identified problem of high prevalence of depression, anxiety, and stress among Sri Lankan university students, coupled with limited accessible, culturally relevant, and proactive support mechanisms.

The main objective was fully achieved: a functional, integrated system was built that predicts severity levels using DASS-21 responses and lifestyle data, and delivers personalized coping suggestions via a mobile application. The specific objectives were met as follows:

- Prevalence and severity of depression, anxiety, and stress were quantified from 1000 USJ student responses, providing updated local insights.
- Lifestyle and campus-environmental factors (sleep, diet, physical activity, ragging perception, etc.) were analyzed and shown to have significant associations with severity levels.
- Machine learning models (LightGBM selected) achieved up to 99% overall exact match accuracy, exceeding expectations and demonstrating reliable prediction.
- Meta-Llama-3.1-8B-Instruct was fine-tuned with LoRA to generate culturally appropriate suggestions based on severity labels and lifestyle profiles.
- A cross-platform Flutter mobile application ("Eternity") was designed, implemented, and integrated with secure backend services (Spring Boot, FastAPI, Firebase), providing real-time screening and multilingual support.

These achievements contribute to early detection, stigma reduction, and self-management support, aligning with national goals for student well-being in higher education.

Limitations

Several limitations impacted the research process and findings:

- The study relied on self-reported data (DASS-21 and lifestyle questions), which may introduce response bias or recall inaccuracy.
- Convenience sampling (1000 responses via Google Forms and campus outreach) may limit generalizability beyond USJ students who were accessible online or physically.

- Cross-sectional design captured data at one point in time, preventing analysis of longitudinal changes or causality between lifestyle factors and mental health.
- LLM fine-tuning was constrained by available hardware (Google Colab T4 GPU) and dataset size, potentially limiting suggestion diversity and depth.
- The prototype was not tested with a large-scale user trial; pilot feedback was limited, so real-world usability and engagement remain to be fully validated.
- Environmental factors (e.g., internet access for real-time inference) and social factors (stigma affecting survey participation) were beyond control.

These limitations are typical for undergraduate applied research and do not undermine the core achievements. They reflect practical constraints of time, resources, and scope.

Future Works / Further Research

The project offers several directions for continuation, either as academic research or commercial development:

- Conduct longitudinal studies to track changes in severity levels and lifestyle factors over time, enabling causal analysis and personalized tracking features.
- Expand user testing with a larger USJ sample and other universities to validate usability (SUS scores), engagement, and effectiveness (pre/post-intervention surveys).
- Integrate clinical validation through partnerships with university counseling or psychiatrists to assess suggestion safety and accuracy.
- Enhance the LLM with a larger, Sri Lanka-specific fine-tuning dataset (local idioms, cultural references) and advanced techniques (e.g., RLHF for empathy).
- Add features like weekly progress tracking, mood logging, or wearable integration (sleep/activity data) for more comprehensive monitoring.
- Explore commercialization: app store release, university licensing, NGO partnerships, or freemium model with premium features (advanced analytics, counselor referral).

These extensions could transform the prototype into a widely adopted digital mental health tool, supporting Sri Lanka's vision for student well-being and innovation in higher education.

In conclusion, this project successfully demonstrated the feasibility of combining psychometric assessment, machine learning prediction, and generative AI in a culturally sensitive mobile

application, offering a practical contribution to mental health support for USJ undergraduates. The work lays a solid foundation for future research and real-world application.

9 .References

- Budiyanto, S., Santoso, H.B. and Hadi, S. (2019) 'Depression and anxiety detection through the closed-loop method using DASS-21', *TELKOMNIKA*, 17(4), pp. 2087–2097. doi: 10.12928/TELKOMNIKA.v17i4.12619.
- Chandramas, K.G.P.R. (no date) *Mellow: Stress management system for university students*. Unpublished thesis. Faculty of Computing, Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka.
- Dhokane, R.M., Wagh, K. and Nikam, A. (2025) 'University student's mental stress detection using machine learning', *International Journal of Scientific Research in Engineering and Management*, 09(03). doi: 10.55041/IJSREM43409.
- Duangchaemkarn, K., Khammarew, P. and Aramvith, S. (2024) 'Machine learning-based classification of mental health state using the DASS-21 profile', in *IEEE Biomedical Engineering International Conference (BMEiCON)*, November 2024, Phayao, Thailand. IEEE.
- Gomes, J.C., De Silva, A.M. and Fernando, S.S. (2024) 'A comparison of mental well-being among female undergraduates in Sri Lanka', *Sri Lanka Journal of Health Sciences*, 4(1). doi: 10.4038/sjhs.v4i1.64.
- Hadzic, M., Vukovic, M. and Stoisavljevic, D. (2024) 'Enhancing early depression detection with AI-powered screening', *SICE Journal of Control, Measurement, and System Integration*, 17(1), pp. 135–143. doi: 10.1080/18824889.2024.2342624.
- Kithmini, J.P.D., Kaarthiga, S., Perera, V.B.J., Deshapriya, H.D.S., Gamage, C.K.W. and Fernando, N.F.J. (2024) 'Depression, anxiety, and stress among allied health sciences undergraduates: A cross-sectional study at a defence university in Sri Lanka', *KDU Journal of Multidisciplinary Studies*, 6(1), pp. 1–12. doi: 10.4038/kjms.v6i1.111.
- Kodagoda, A.S.K. and Meegoda, M.K.D.L. (2020) 'The prevalence of depression among undergraduates of a selected medical faculty in Sri Lanka', *Journal of Health Sciences and Innovative Research*, 1(1), pp. 22–32.

- Kruchten, P. (1995) 'The 4+1 view model of architecture', *IEEE Software*, 12(6), pp. 42–50. doi: 10.1109/52.469759.
- Lovibond, S.H. and Lovibond, P.F. (1995) *Manual for the depression anxiety stress scales*. 2nd ed. Sydney: Psychology Foundation of Australia.
- Madububambachu, U., Ihezue, U., Ukpebor, A. and Ayo, S. (2024a) 'Predictive machine learning approaches for mental health diagnoses in college students', *International Journal on Engineering Technologies and Informatics*, 5(3), pp. 1–10.
- Madububambachu, U., Ukpebor, A. and Ihezue, U. (2024b) 'Machine learning techniques to predict mental health diagnoses: A systematic literature review', *Clinical Practice & Epidemiology in Mental Health*, 20, pp. 1–15. doi: 10.2174/174501790242001.
- Mandal, S., Singh, P. and Gupta, R. (2025) 'Stress detection using wearable devices and machine learning', *Procedia Computer Science*, 250, pp. 749–759. doi: 10.1016/j.procs.2025.01.069.
- Mutalib, S., Shafiee, N.S.M. and Abdul-Rahman, S. (2021) 'Mental health prediction models using machine learning in higher education institution', *Turkish Journal of Computer and Mathematics Education*, 12(5), pp. 1–9.
- Pathak, V., Kumar, R. and Sharma, S. (2024) 'Mental health prediction using modified DASS-21 based on ensemble machine learning', *TEJAS Journal*, 3(3). doi: 10.5281/zenodo.13844425.
- Rahman, H., Kwicklis, M., Ottom, M., Amornsriwatanakul, A., Abdul-Mumin, K.H., Rosenberg, M. and Dinov, I.D. (2023) 'Machine learning-based prediction of mental well-being using health behavior data from university students', *Bioengineering*, 10(5), 575. doi: 10.3390/bioengineering10050575.
- Sajja, G. (2021) 'Machine learning based detection of depression and anxiety using speech analysis', *International Journal of Computer Applications*, 183(45), pp. 10–16.
- Valinskas, S., Nakas, M., Petrenas, A., Marozas, V. and Bacevicius, J. (2023) 'Sensa mobile app for managing stress, anxiety, and depression: A mixed-methods feasibility study', *JMIR Formative Research*, 7, e40671. doi: 10.2196/40671.
- Vitória, A.R.S., Brandão, T., Pimentel, M.L. and Costa, R.M.D. (2024) 'A machine learning approach for mental health monitoring', *FedCSIS (Communication Papers)*, pp. 187–194. doi: 10.15439/2024F3586.

- Wickramasinghe, A., Wijenayake, S. and Perera, D. (2023) 'Prevalence of depression among students at a Sri Lankan university during the COVID-19 pandemic: A cross-sectional study', *BMC Public Health*, 23, 528. doi: 10.1186/s12889-023-15427-y.
- Wijesekara, P.A.D.S.N. (2022) 'A study in University of Ruhuna for investigating prevalence, risk factors and remedies for psychiatric illnesses among students', *Scientific Reports*, 12, 12763. doi: 10.1038/s41598-022-17111-6.